

Selection Bias and Imputation in the Potential Outcomes Framework

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1. The potential outcomes setup: what could have been

Think about a group of people, indexed by $i = 1, \dots, n$. Each person either receives treatment ($X_i = 1$, like drinking coffee) or doesn't ($X_i = 0$). Here's the key insight: we can imagine *two parallel universes* for each person. In one universe they get treatment and we observe outcome $Y_i(1)$; in the other they don't and we observe $Y_i(0)$. These are the *potential outcomes*.¹

Of course, we only live in one universe. The outcome we actually see follows the *consistency rule*:

$$Y_i^{\text{obs}} = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

This just says: if you took treatment ($X_i = 1$), we observe $Y_i(1)$; otherwise we observe $Y_i(0)$.

The causal effect for person i is simply the difference between these two potential worlds: $\tau_i = Y_i(1) - Y_i(0)$. When we talk about average effects, two quantities matter most:

- The *average treatment effect* (ATE): $\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$, averaging over everyone.
- The *average treatment effect on the treated* (ATT): $\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = 1]$, averaging only over those who actually got treated.

Throughout, we assume *SUTVA* (stable unit treatment value assumption)²: your potential outcomes don't change depending on whether your neighbor gets treatment, and there's only one version of the treatment.

*The structure and some references in this handout are adapted from Zed Hoffman-Weldon (2025), "POLI110: The Basic Theory of Experiments," with additions for selection bias and imputation. All errors are my own.

¹Jerzy Neyman, "On the Application of Probability Theory to Agricultural Experiments," *Statistical Science* 5 (1990): 465–480 (Reprint of 1923 article); Donald B. Rubin, "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies," *Journal of Educational Psychology* 66, no. 5 (1974): 688–701; Guido W. Imbens and Donald B. Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences* (Cambridge, UK: Cambridge University Press, 2015).

²Rubin, "Estimating Causal Effects of Treatments"; Imbens and Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences*.

When we run experiments, it helps to distinguish assignment from actual receipt. Let $Z_i \in \{0, 1\}$ be who the experimenter *assigns* to treatment (randomly), and X_i be who *actually takes* it. With perfect compliance, $Z_i = X_i$. But in the real world people don't always follow instructions, so we might study the *intent-to-treat* effect (effect of assignment Z on outcomes) or, with more assumptions, the effect among *compliers*.

2. Why naive comparisons mislead: the selection bias decomposition

Here's the fundamental problem: we can't see both $Y_i(1)$ and $Y_i(0)$ for the same person. So a natural instinct is to compare the average outcome among the treated to the average among the untreated: $\mathbb{E}[Y_i | X_i = 1] - \mathbb{E}[Y_i | X_i = 0]$.

But this comparison mixes two things together. Using a bit of algebra (adding and subtracting the same term), we can write:³

$$\begin{aligned} \mathbb{E}[Y_i | X_i = 1] - \mathbb{E}[Y_i | X_i = 0] &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) | X_i = 1]}_{\text{ATT: the real effect}} \\ &\quad + \underbrace{\mathbb{E}[Y_i(0) | X_i = 1] - \mathbb{E}[Y_i(0) | X_i = 0]}_{\text{selection bias}}. \end{aligned}$$

Let's unpack this. The first term is what we want: the actual causal effect among those who were treated. The second term is the problem: it's the difference in *baseline* outcomes (in the untreated state) between those who ended up treated and those who didn't. If people who choose coffee would have had different stress levels even *without* coffee, then comparing their outcomes to non-coffee-drinkers conflates the treatment effect with these pre-existing differences.

Figure 1 illustrates this problem with a directed acyclic graph (DAG).⁴ An unobserved confounder U (like "baseline stress predisposition") affects both who chooses treatment (X) and the outcome (Y), creating a backdoor path between X and Y that isn't causal.

3. The coffee-and-stress story: from three people to many

Let's revisit the classroom example with three people: Alice and Carol drink coffee; Bob doesn't. Here's what we know (including the unobserved counterfactuals, which in reality we never see):

³Paul W. Holland, "Statistics and Causal Inference," *Journal of the American Statistical Association* 81, no. 396 (1986): 945–960; Donald B. Rubin, "Causal Inference Using Potential Outcomes: Design, Modeling, Decisions," *Journal of the American Statistical Association* 100, no. 469 (2005): 322–331.

⁴Judea Pearl, *Causality*, 2nd ed. (2000; Cambridge, UK: Cambridge University Press, 2009).

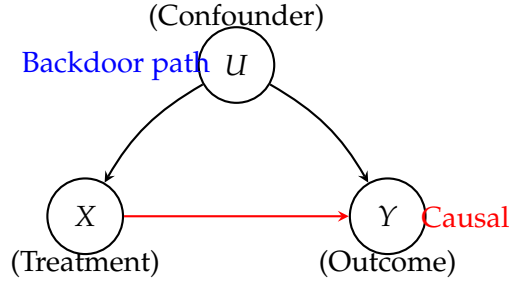


Figure 1: Selection bias arises when unobserved factors U influence both treatment choice X and outcomes Y , creating a non-causal association.

Person	$Y_i(1)$ (Drink)	$Y_i(0)$ (No drink)	τ_i	Observed Y_i^{obs}
Alice	8	4	+4	8 (drinks)
Bob	7	6	+1	6 (no drink)
Carol	9	3	+6	9 (drinks)

The true ATT is the average effect among drinkers: $\text{ATT} = [(8 - 4) + (9 - 3)]/2 = 5$. Coffee really does increase stress by 5 points, on average, for those who drink it.

Now suppose we naively compare mean outcomes: drinkers average $(8 + 9)/2 = 8.5$, Bob (the only non-drinker) has 6, so the naive difference is $8.5 - 6 = 2.5$. This *underestimates* the true effect!

Why? Look at the baseline (no-coffee) potential outcomes: Alice and Carol would have averaged $(4 + 3)/2 = 3.5$ without coffee, while Bob's baseline is 6. The selection bias is $3.5 - 6 = -2.5$. The people who chose coffee would have been *less stressed* even without coffee. The naive comparison mixes the true effect (+5) with this baseline difference (-2.5), giving us $2.5 = 5 + (-2.5)$.

Scaling up. Now imagine $n = 100$ people: 60 drink coffee, 40 don't. Suppose among coffee drinkers, the average untreated stress is $\overline{Y(0) \mid X = 1} = 6.0$ and treated stress is $\overline{Y(1) \mid X = 1} = 8.0$, so $\text{ATT} = 2.0$. Among non-drinkers, average stress is $\overline{Y(0) \mid X = 0} = 5.0$. The naive comparison gives $8.0 - 5.0 = 3.0$, but the selection bias is $6.0 - 5.0 = 1.0$, so we're overstating the effect: $3.0 = 2.0 + 1.0$.

Here's the crucial lesson: *collecting more data doesn't fix this*. If we survey 1,000 or 10,000 people, the sample means will converge to these population means by the law of large numbers⁵—but the bias remains. We'll just estimate the *wrong thing* more precisely. Bigger samples give us tighter confidence intervals around a biased estimate. That's not progress.

⁵Rick Durrett, *Probability: Theory and Examples*, 5th ed., 2019; A. W. van der Vaart, *Asymptotic Statistics* (Cambridge, UK: Cambridge University Press, 1998).

4. How randomization saves the day

In a proper randomized controlled trial (RCT),⁶ we assign treatment randomly: $X_i \perp \{Y_i(0), Y_i(1)\}$. This independence means that, on average, the treated and control groups are identical in *all respects*—observed and unobserved—except for the treatment itself. So $\mathbb{E}[Y_i(0) \mid X_i = 1] = \mathbb{E}[Y_i(0) \mid X_i = 0]$, and the selection bias term vanishes.

Figure 2 shows the difference: when X is randomized, there are no arrows pointing into it, so no backdoor paths exist.

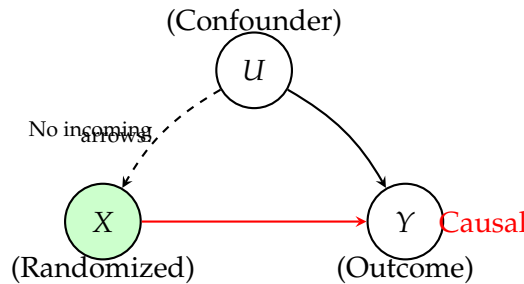


Figure 2: Under randomization, X is independent of all potential confounders, eliminating selection bias.

The difference in means between treatment and control groups becomes an unbiased estimator of the causal effect. As n grows, the law of large numbers ensures we estimate that effect more and more precisely. This is why experiments are considered the gold standard.

With non-compliance ($Z \neq X$), the difference in means by assignment Z identifies the intent-to-treat effect. With additional assumptions (monotonicity, exclusion restrictions), instrumental variables logic lets us recover the effect for compliers.

5. The imputation perspective: filling in what's missing

Here's a unifying way to think about causal estimation: we're always trying to "fill in" the missing potential outcome. For ATT, we need to compare $Y_i(1)$ to $Y_i(0)$ for the treated units, but we only observe $Y_i(1)$. So we *impute* $\widehat{Y_i(0)}$ using observable information:

$$\text{ATT} \approx \frac{1}{|T|} \sum_{i \in T} (Y_i^{\text{obs}} - \widehat{Y_i(0)}),$$

where $T = \{i : X_i = 1\}$ is the treated group and $\widehat{Y_i(0)}$ is our best guess at what their outcome would have been without treatment.

Different methods correspond to different imputation strategies:

⁶Alan S. Gerber and Donald P. Green, *Field Experiments: Design, Analysis, and Interpretation* (New York, NY: W. W. Norton, 2012).

- **RCT:** Impute with the control group mean. Randomization guarantees this is unbiased.
- **Regression:** Estimate $m_0(z) = \mathbb{E}[Y \mid X = 0, Z = z]$ from control units as a function of covariates Z , then set $\widehat{Y_i(0)} = m_0(Z_i)$ for treated units.⁷
- **Matching:** Find control units similar to each treated unit (based on covariates) and use their observed outcomes as $\widehat{Y_i(0)}$.⁸
- **IPW (inverse probability weighting):** Reweight controls to match the treated group's covariate distribution, then average.⁹
- **Doubly robust methods:** Combine regression and weighting so the estimator is consistent if *either* model is correct.¹⁰
- **Difference-in-differences:** Impute the treated group's post-treatment counterfactual by adding the control group's trend to the treated group's pre-treatment level, valid under parallel trends.¹¹

Thinking about imputation clarifies what assumptions each method requires and where they might fail.

6. A classroom-friendly larger- n example

For a board demonstration, try this: $n = 10$, with 6 treated and 4 controls. Among the treated, suppose their untreated potential outcomes average 5.5, and they have individual effects of (1, 2, 2, 3, 3, 4). Then their treated outcomes average $5.5 + 2.5 = 8.0$ (since $ATT = 2.5$). Wait, let me recalculate: if effects are (1, 2, 2, 3, 3, 4), the average is $(1 + 2 + 2 + 3 + 3 + 4)/6 = 15/6 = 2.5$. So treated mean is $5.5 + 2.5 = 8.0$.

Among controls, the untreated mean is 4.5. Then:

- $ATT = 8.0 - 5.5 = 2.5$
- Selection bias = $5.5 - 4.5 = 1.0$
- Naive difference = $8.0 - 4.5 = 3.5$

And indeed, $3.5 = 2.5 + 1.0$.

If you randomly reassign who's treated while keeping potential outcomes fixed, selection bias averages to zero. If you keep the self-selection pattern fixed and just resample, the bias persists.

⁷Joshua Angrist and Jörn-Steffen Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion* (Princeton, NJ: Princeton University Press, 2008); Miguel A. Hernán and James M. Robins, *Causal Inference: What If* (Chapman & Hall/CRC, 2020).

⁸Paul R. Rosenbaum, *Observational Studies*, 2nd ed. (Springer, 2002).

⁹Paul R. Rosenbaum and Donald B. Rubin, "The central role of the propensity score in observational studies for causal effects," *Biometrika* 70, no. 1 (1983): 41–55; Hernán and Robins, *Causal Inference: What If*.

¹⁰Hernán and Robins, *Causal Inference: What If*.

¹¹Angrist and Pischke, *Mostly Harmless Econometrics*.

Appendix A: Mathematical derivations

A.1 Proof of the selection bias decomposition

We want to show:

$$\mathbb{E}[Y_i \mid X_i = 1] - \mathbb{E}[Y_i \mid X_i = 0] = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = 1] + \mathbb{E}[Y_i(0) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 0].$$

Proof. Start with the left-hand side and use the consistency rule. For the treated group:

$$\mathbb{E}[Y_i \mid X_i = 1] = \mathbb{E}[Y_i(1) \mid X_i = 1],$$

since when $X_i = 1$, we observe $Y_i(1)$. For the control group:

$$\mathbb{E}[Y_i \mid X_i = 0] = \mathbb{E}[Y_i(0) \mid X_i = 0].$$

So:

$$\mathbb{E}[Y_i \mid X_i = 1] - \mathbb{E}[Y_i \mid X_i = 0] = \mathbb{E}[Y_i(1) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 0].$$

Now add and subtract $\mathbb{E}[Y_i(0) \mid X_i = 1]$:

$$\begin{aligned} &= \mathbb{E}[Y_i(1) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 1] + \mathbb{E}[Y_i(0) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 0] \\ &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 0]}_{\text{selection bias}}, \end{aligned}$$

by linearity of expectation. □

A.2 Why randomization makes selection bias zero

Proof. Under randomization, $X_i \perp \{Y_i(0), Y_i(1)\}$. By definition of independence:

$$\mathbb{E}[Y_i(0) \mid X_i = 1] = \mathbb{E}[Y_i(0)] = \mathbb{E}[Y_i(0) \mid X_i = 0].$$

Therefore:

$$\text{Selection bias} = \mathbb{E}[Y_i(0) \mid X_i = 1] - \mathbb{E}[Y_i(0) \mid X_i = 0] = 0.$$

Similarly, under randomization:

$$\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = 1] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] = \text{ATE}.$$

So the naive difference in means equals the ATE:

$$\mathbb{E}[Y_i \mid X_i = 1] - \mathbb{E}[Y_i \mid X_i = 0] = \text{ATE}.$$

□

A.3 Sample means and the law of large numbers

Let $\bar{Y}_T = \frac{1}{n_T} \sum_{i: X_i=1} Y_i$ be the sample mean among treated units, where $n_T = \sum_i X_i$ is the number treated. By the weak law of large numbers, as $n_T \rightarrow \infty$:

$$\bar{Y}_T \xrightarrow{p} \mathbb{E}[Y_i \mid X_i = 1],$$

where $\xrightarrow{p}$ denotes convergence in probability. Similarly for the control group.

This means that with large samples, we precisely estimate the conditional means $\mathbb{E}[Y \mid X = 1]$ and $\mathbb{E}[Y \mid X = 0]$. But if there's selection bias—if $\mathbb{E}[Y(0) \mid X = 1] \neq \mathbb{E}[Y(0) \mid X = 0]$ —then the difference in these conditional means does *not* equal the causal effect, no matter how large n is.

Formally, even as $n \rightarrow \infty$:

$$\bar{Y}_T - \bar{Y}_C \xrightarrow{p} \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0] = \text{ATT} + \text{Selection Bias}.$$

If selection bias $\neq 0$, we converge to the wrong quantity. We need an *identification strategy* (like randomization, or parallel trends, or unconfoundedness), not just more data.

A.4 Variance of the difference-in-means estimator

Under randomization with n_T treated and $n_C = n - n_T$ controls, the variance of the difference-in-means estimator is:

$$\text{Var}(\bar{Y}_T - \bar{Y}_C) = \frac{\sigma_T^2}{n_T} + \frac{\sigma_C^2}{n_C},$$

where $\sigma_T^2 = \text{Var}(Y_i \mid X_i = 1)$ and $\sigma_C^2 = \text{Var}(Y_i \mid X_i = 0)$.

As $n \rightarrow \infty$ with fixed treatment probability $p = n_T/n$, this variance goes to zero, so our estimate converges to the true ATE. The standard error is approximately:

$$\text{SE}(\bar{Y}_T - \bar{Y}_C) \approx \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}},$$

where s_T^2 and s_C^2 are sample variances.

Appendix B: Extended coffee-and-stress calculations

Example 1 (Small sample: $n = 10$). Suppose we have 10 people. Six drink coffee ($X = 1$), four don't ($X = 0$).

Potential outcomes:

Person	X_i	$Y_i(0)$	$Y_i(1)$	τ_i	Y_i^{obs}
1	1	5	6	1	6
2	1	5	7	2	7
3	1	5	7	2	7
4	1	6	9	3	9
5	1	6	9	3	9
6	1	6	10	4	10
7	0	4	5	1	4
8	0	4	6	2	4
9	0	5	7	2	5
10	0	5	8	3	5

Calculations:

- Treated group: $\overline{Y(0) \mid X = 1} = (5 + 5 + 5 + 6 + 6 + 6)/6 = 5.5$
- Treated group: $\overline{Y(1) \mid X = 1} = (6 + 7 + 7 + 9 + 9 + 10)/6 = 8.0$
- Control group: $\overline{Y(0) \mid X = 0} = (4 + 4 + 5 + 5)/4 = 4.5$
- $\text{ATT} = 8.0 - 5.5 = 2.5$
- Selection bias = $5.5 - 4.5 = 1.0$
- Naive difference = $8.0 - 4.5 = 3.5$

Verify: $3.5 = 2.5 + 1.0$.

The people who chose coffee would have had stress level 5.5 on average even without coffee, compared to 4.5 for non-coffee-drinkers. This +1.0 baseline difference inflates the naive estimate from the true ATT of 2.5 to an observed difference of 3.5.

Example 2 (Moderate sample: $n = 100$). Suppose 60 drink coffee, 40 don't. Population parameters:

- $\mathbb{E}[Y(0) \mid X = 1] = 6.0, \mathbb{E}[Y(1) \mid X = 1] = 8.0 \Rightarrow \text{ATT} = 2.0$
- $\mathbb{E}[Y(0) \mid X = 0] = 5.0$

Then:

- Selection bias = $6.0 - 5.0 = 1.0$
- Naive difference = $8.0 - 5.0 = 3.0$
- Decomposition: $3.0 = 2.0 + 1.0$

As n increases from 100 to 1000 to 10,000, the sampling variability around these means

decreases, but the bias remains. With $n = 10,000$:

$$\text{SE}(\bar{Y}_T - \bar{Y}_C) \approx \sqrt{\frac{\sigma^2}{6000} + \frac{\sigma^2}{4000}} \propto \frac{1}{\sqrt{n}} \rightarrow 0,$$

but we still converge to 3.0, not 2.0.

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